

Week 7

Student Performance – Data Cleaning & Transformation

(Power BI Desktop)

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1. Aim

The aim of this week is to take the Student Performance in Exams dataset and clean it, fix the messy parts, add new calculated columns, reshape the marks data, and build a small data model with some basic DAX measures so that it is ready for a dashboard in Week 8.

2. Dataset

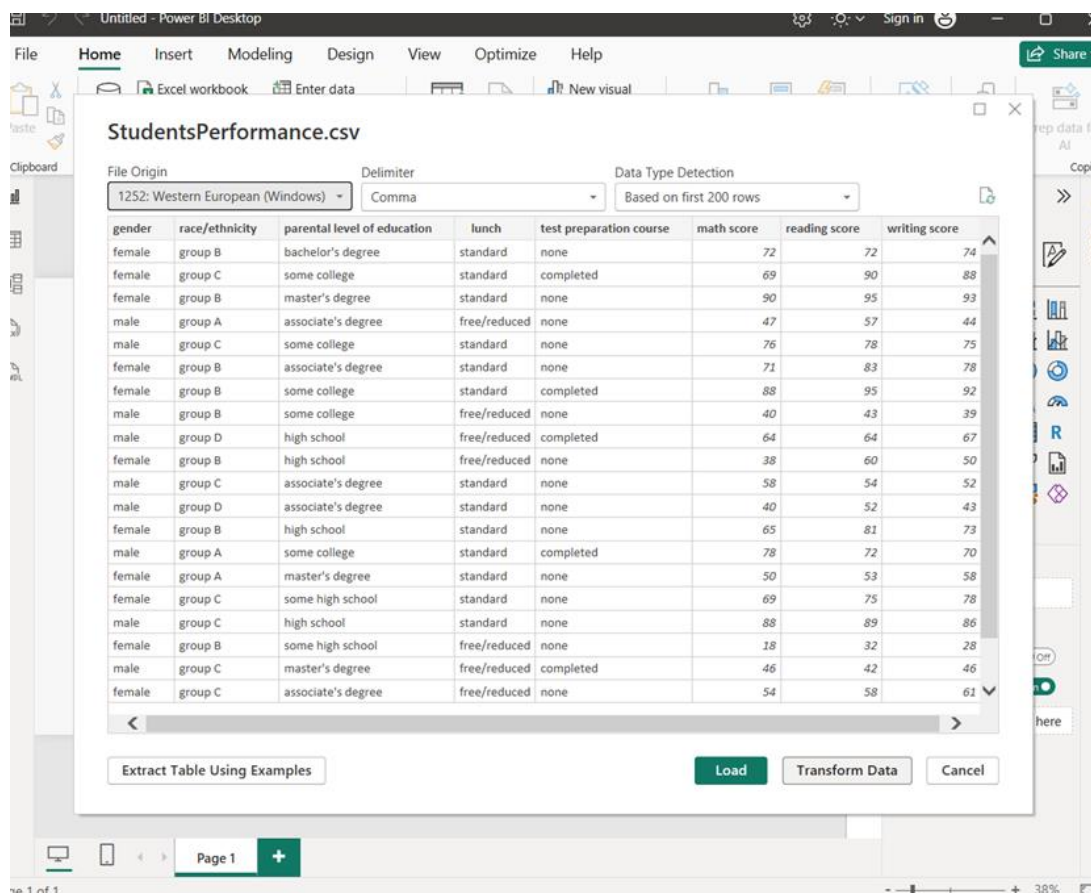
I used the Student Performance in Exams dataset (StudentsPerformance.csv). It has 1000 students and these columns:

- Categorical columns: gender, race/ethnicity, parental level of education, lunch, test preparation course.
- Numerical columns: math score, reading score, writing score.

3. Procedure

Step 1: Importing the data

I opened Power BI Desktop and used Get Data → Text/CSV to load StudentsPerformance.csv. In the preview window I checked that the delimiter was comma and then clicked Transform Data to open Power Query Editor.

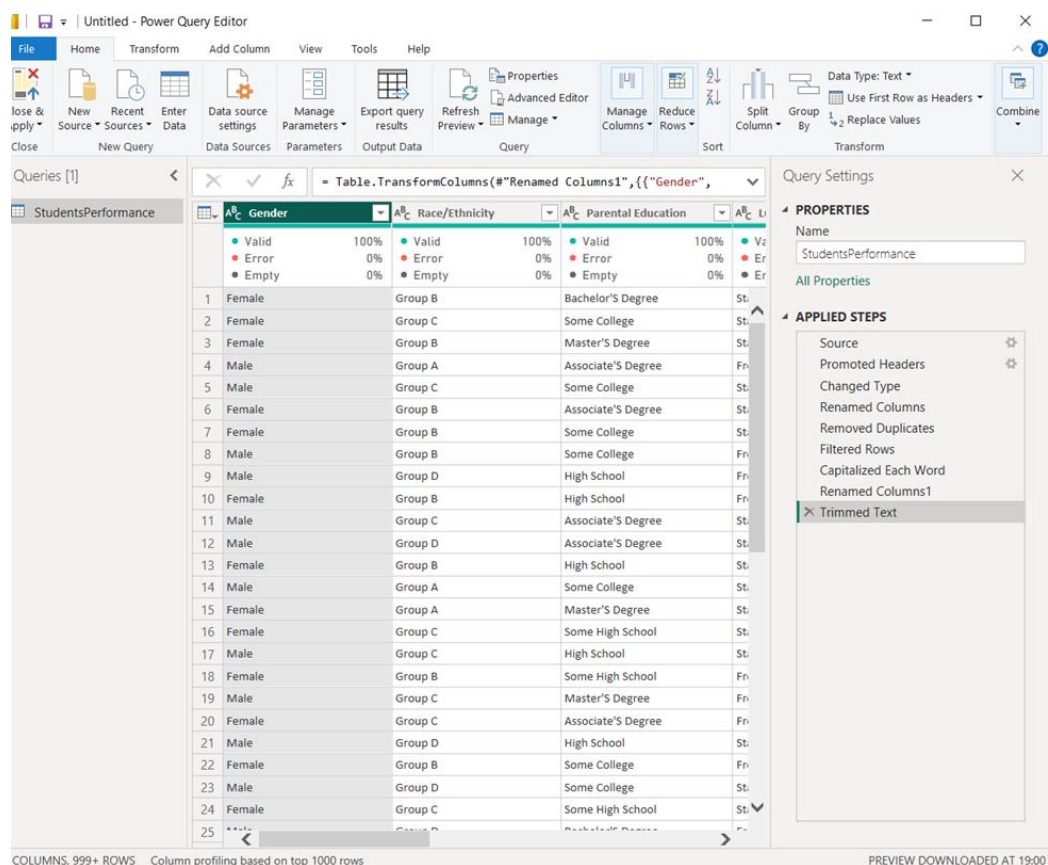


Step 2 to 4: Exploring and cleaning the data

First I looked through the columns and rows and checked the data types. Then I cleaned the data in Power Query. The steps I applied are:

- Promoted the first row to headers and set the correct data types.
- Renamed the columns so they are clear (for example “gender” became “Gender” and “parental level of education” became “Parental Education”).
- Removed duplicate rows.
- Filtered out the blank or invalid rows.
- Used Capitalize Each Word so all the text values look the same (Female, Group B, Some College and so on).
- Used Trim to remove extra spaces from the text columns.

All of these steps can be seen in the Applied Steps list on the right side of the screen.



Step 5: Creating calculated columns

I added new columns using Add Column → Custom Column.

- Total Score = Math Score + Reading Score + Writing Score
- Average Score = Total Score / 3
- Performance Grade = a conditional column that gives A, B, C or F depending on the average score (A is high performance, B is good, C is average and F is low).

Custom Column

Add a column that is computed from the other columns.

New column name: Total Score

Custom column formula: = [Math Score] + [Reading Score] + [Writing Score]

Available columns: Gender, Race/Ethnicity, Parental Education, Lunch, Test Preparation Course, Math Score, Reading Score, Writing Score

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✓ No syntax errors have been detected.

OK Cancel

	Lunch	Math Score	Reading Score	Writing Score	Total Score
1	Standard				
2	Standard				
3	Standard				
4	Free/Reduced				
5	Standard				
6	Standard				
7	Standard				
8	Free/Reduced				
9	Free/Reduced				
10	Free/Reduced				
11	Standard				
12	Standard				
13	Standard				
14	Standard				
15	Standard				
16	Standard				
17	Standard				
18	Free/Reduced				
19	Free/Reduced	Completed	46	42	46
20	Free/Reduced	None	54	58	61
21	Standard	None	66	69	63
22	Free/Reduced	Completed	65	75	70
23	Standard	None	44	54	53
24	Standard	None	69	73	73
25	Free/Reduced	Completed	74	71	80

Step 6: Pass/Fail classification

I took the pass mark as an average score of 40 or more. Using Add Column → Conditional Column I wrote: if Average Score $\geq$ 40 then “Pass” else “Fail”. For example, the student with an average of 26 gets F and Fail, while students with averages above 40 get Pass.

Pass/Fail

Add a column that is computed from the other columns.

New column name: Pass/Fail

Custom column formula: = if [Average Score] $\geq$ 40 then "Pass" else "Fail"

Available columns: Reading Score, Writing Score, Total Score, Average Score, Performance Grade, Pass/Fail

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Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

	Reading Score	Writing Score	Total Score	Average Score	Performance Grade	Pass/Fail	
1	72	72	74	218	72.66666667	B	Pass
2	69	90	88	247	82.33333333	A	Pass
3	90	95	93	278	92.66666667	A	Pass
4	47	57	44	148	49.33333333	C	Pass
5	76	78	75	229	76.33333333	B	Pass
6	71	83	78	232	77.33333333	B	Pass
7	88	95	92	275	91.66666667	A	Pass
8	40	43	39	122	40.66666667	C	Pass
9	64	64	67	195	65	B	Pass
10	38	60	50	148	49.33333333	C	Pass
11	58	54	52	164	54.66666667	C	Pass
12	40	52	43	135	45	C	Pass
13	65	81	73	219	73	B	Pass
14	78	72	70	220	73.33333333	B	Pass
15	50	53	58	161	53.66666667	C	Pass
16	69	75	78	222	74	B	Pass
17	88	89	86	263	87.66666667	A	Pass
18	18	32	28	78	26	F	Fail
19	46	42	46	134	44.66666667	C	Pass
20	54	58	61	173	57.66666667	C	Pass
21	66	69	63	198	66	B	Pass
22	65	75	70	210	70	B	Pass
23	44	54	53	151	50.33333333	C	Pass

Step 7: Reshaping the subject data (Unpivot)

Math, Reading and Writing were three separate columns, which makes it hard to compare subjects. So I selected these three columns and used Transform → Unpivot Only Selected Columns. After that I renamed the new columns to Subject and Score. Now every student has three rows, one for each subject, so subjects can be compared easily.

Table: StudentsPerformance

Course	Reading Score	Writing Score	Total Score
1	72	72	74
2	69	90	88
3	90	95	93
4	47	57	44
5	76	78	75
6	71	83	78
7	88	95	92
8	40	43	39
9	64	64	67
10	38	60	50
11	58	54	52
12	40	52	43
13	65	81	73
14	78	72	70
15	50	53	58
16	69	75	78
17	88	89	86
18	18	32	28
19	46	42	46
20	54	58	61
21	66	69	63
22	65	75	70
23	44	54	53
24	69	73	73
25	74	71	80
26	73	74	72
27	69	54	55

Table: StudentsPerformance

Course	Total Score	Average Score	Performance Grade	Pass/Fail	Subject	Score
1	218	72.66666667	B	Pass	Math Score	72
2	218	72.66666667	B	Pass	Reading Score	72
3	218	72.66666667	B	Pass	Writing Score	74
4	247	82.33333333	A	Pass	Math Score	69
5	247	82.33333333	A	Pass	Reading Score	90
6	247	82.33333333	A	Pass	Writing Score	88
7	278	92.66666667	A	Pass	Math Score	90
8	278	92.66666667	A	Pass	Reading Score	95
9	278	92.66666667	A	Pass	Writing Score	93
10	148	49.33333333	C	Pass	Math Score	47
11	148	49.33333333	C	Pass	Reading Score	57
12	148	49.33333333	C	Pass	Writing Score	44
13	229	76.33333333	B	Pass	Math Score	76
14	229	76.33333333	B	Pass	Reading Score	78
15	229	76.33333333	B	Pass	Writing Score	75
16	232	77.33333333	B	Pass	Math Score	71
17	232	77.33333333	B	Pass	Reading Score	83
18	232	77.33333333	B	Pass	Writing Score	78
19	275	91.66666667	A	Pass	Math Score	88
20	275	91.66666667	A	Pass	Reading Score	95
21	275	91.66666667	A	Pass	Writing Score	92
22	122	40.66666667	C	Pass	Math Score	40

Step 8: Preparing the data model

Next I created extra tables from the main query so the model is neat and organised. I used the Reference/Duplicate option and removed the columns that were not needed in each table.

- Student Base: the cleaned table with all the columns, including the calculated ones.
- Students Details: one row per student with details like Gender, Lunch, Test Preparation Course, Total Score, Average Score, Performance Grade and Pass/Fail (I removed the three score columns from here).

- Students Marks: Student ID, Subject and Score after unpivoting the three score columns.

Power BI Desktop interface showing the 'StudentsPerformance' table with columns: Student ID, Writing Score, Reading Score, Math Score, and Average Score. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' for the 'StudentsPerformance' query, including 'Added Conditional Column1'.

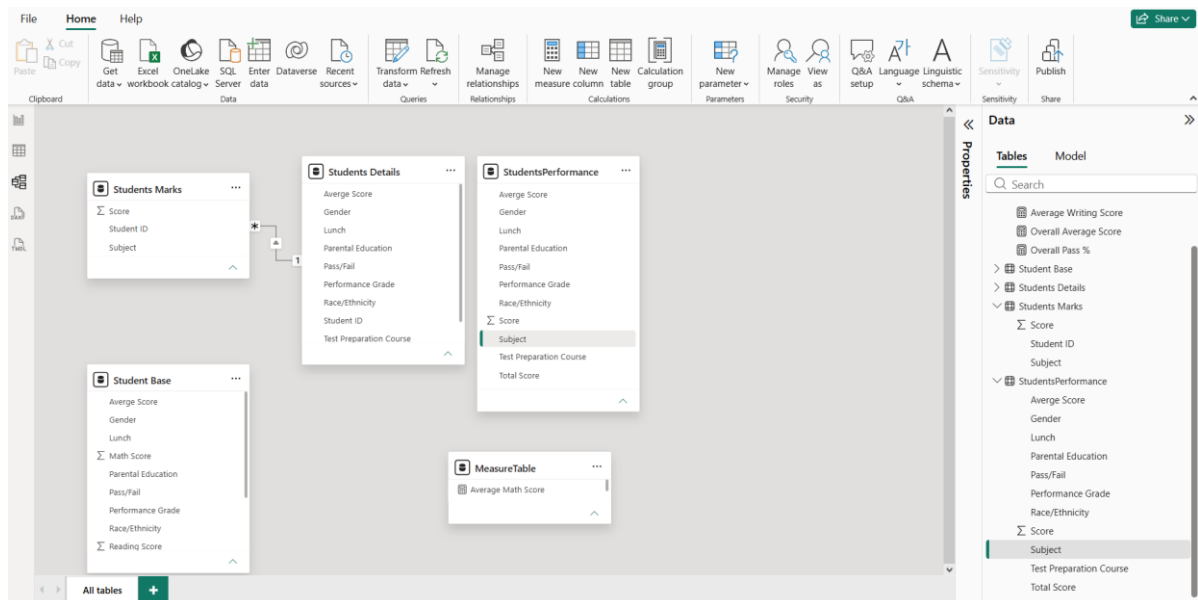
Student ID	Writing Score	Reading Score	Math Score	Average Score
1	72	74	218	72.66666667
2	90	88	247	82.33333333
3	95	93	278	92.66666667
4	57	44	148	49.33333333
5	78	75	229	76.33333333
6	83	78	232	77.33333333
7	95	92	275	91.66666667
8	43	39	122	40.66666667
9	64	67	195	65.66666667
10	60	50	148	49.33333333
11	54	52	164	54.66666667
12	52	43	135	45.66666667
13	81	73	219	73.66666667
14	72	70	220	73.33333333
15	53	58	161	53.66666667
16	75	78	222	74.66666667
17	89	86	263	87.66666667
18	32	28	78	26.66666667
19	42	46	134	44.66666667
20	58	61	173	57.66666667
21	69	63	198	66.66666667
22	75	70	210	70.66666667
23	54	53	151	50.66666667
24	73	73	215	71.66666667
25	77	77	235	77.66666667

Power BI Desktop interface showing the 'StudentsMarks' table with columns: Student ID, Subject, Score, and Pass/Fail. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' for the 'StudentsMarks' query, including 'Removed Columns'.

Student ID	Subject	Score	Pass/Fail
1	Standard	None	Pass
2	Standard	Completed	Pass
3	Standard	None	Pass
4	Free/Reduced	None	Pass
5	Standard	None	Pass
6	Standard	None	Pass
7	Standard	Completed	Pass
8	Free/Reduced	None	Pass
9	Free/Reduced	Completed	Pass
10	Free/Reduced	None	Pass
11	Standard	None	Pass
12	Standard	None	Pass
13	Standard	None	Pass
14	Standard	Completed	Pass
15	Standard	None	Pass
16	Standard	None	Pass
17	Standard	None	Pass
18	Free/Reduced	None	Fail
19	Free/Reduced	Completed	Pass
20	Free/Reduced	None	Pass
21	Standard	None	Pass
22	Free/Reduced	Completed	Pass
23	Standard	None	Pass
24	Standard	None	Pass
25	Free/Reduced	Completed	Pass

Student ID	Subject	Score
1	Math Score	72
2	Reading Score	72
3	Writing Score	74
4	Math Score	69
5	Reading Score	90
6	Writing Score	88
7	Math Score	90
8	Reading Score	90
9	Writing Score	93
10	Math Score	47
11	Reading Score	57
12	Writing Score	44
13	Math Score	76
14	Reading Score	78
15	Writing Score	75
16	Math Score	71
17	Reading Score	83
18	Writing Score	78
19	Math Score	88
20	Reading Score	95
21	Writing Score	92
22	Math Score	40
23	Reading Score	43
24	Writing Score	39
25	Math Score	64
26	Reading Score	64

Student ID is the common key between Students Details and Students Marks. In the Model view I connected them with a one-to-many relationship: one student in Students Details can have many rows (three subjects) in Students Marks.

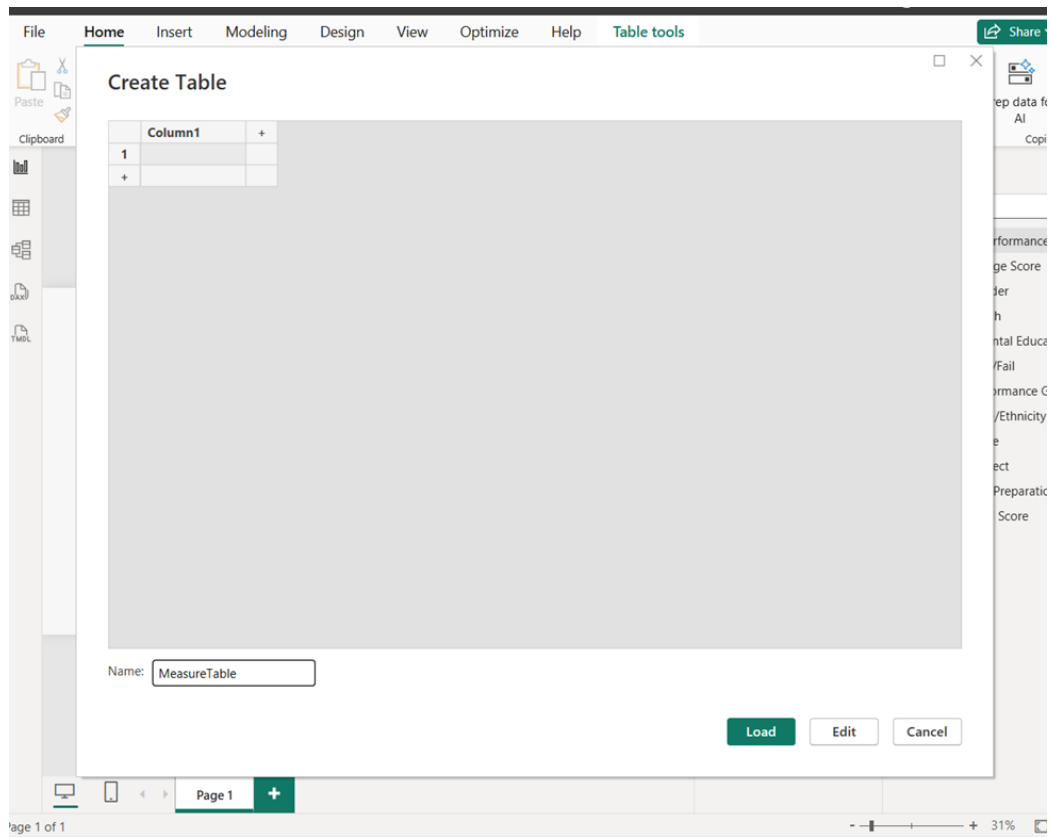


Step 9: Loading the data

I clicked Close & Apply to load all the tables into Power BI, and then checked the tables in Data view to make sure the new columns and the unpivoted data looked correct.

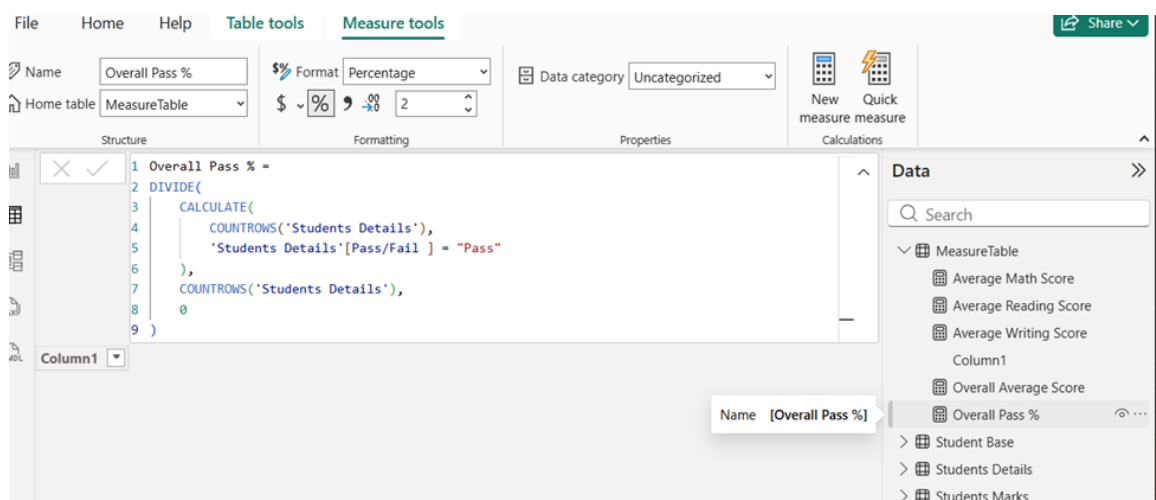
Step 10: Creating DAX measures

I made a new table called MeasureTable using Enter Data and kept all my measures in it so they are easy to find.



The measures I created are:

- Average Math Score
- Average Reading Score
- Average Writing Score
- Overall Average Score
- Overall Pass %



4. Validation

To check that everything was correct, I compared a few rows with the original dataset.

- The first student has Math 72, Reading 72 and Writing 74. The total is 218 and the average is 72.67, which matches the Total Score and Average Score columns, and the grade is B with Pass.
- One student scored 18, 32 and 28. The total is 78 and the average is 26, so the grade is F and the result is Fail, which is correct.
- After unpivoting, each student has three rows (Math, Reading, Writing) and the scores match the original columns.

So the calculated columns, the Pass/Fail classification, the grades and the unpivoted data are all working properly.

5. Output

At the end of Week 7 I have a cleaned and structured Student Performance dataset with calculated columns (Total Score, Average Score, Performance Grade, Pass/Fail), unpivoted subject data in Subject and Score columns, separate Student Base, Students Details and Students Marks tables with a relationship on Student ID, and basic DAX measures in MeasureTable.

6. Conclusion

In this week I learned how to clean and transform data using Power Query, how to create calculated and conditional columns, how unpivot makes subject-wise analysis easier, and how to write simple DAX measures. The data model is now ready for building the dashboard in Week 8.